ABSTRACT

Electrocardiogram(ECG) analysis plays a vital role in the early detection of cardiovascular diseases,particularly arrhythmias,which can lead to severe health complications if not diagnosed in time.This project presents a hybrid deep learning approach for automated classification of heartbeat signals using the MIT-BIH Arrhythmia dataset.The proposed model integrates Convolutional Neural Networks (CNN) for extracting local morphological features from ECG waveforms, BR-SquareNet residual blocks for refining features and ensuring stable deep learning, and Bidirectional Long Short-Term Memory (BiLSTM) networks for capturing temporal dependencies and rhythm patterns in sequential cardiac signals.

The ECG data is preprocessed and structured into a format suitable for one-dimensional convolutional analysis. The CNN layers learn hierarchical waveform characteristics, while residual connections preserve original signal information and enhance subtle anomaly detection. The BiLSTM layer further models time-based variations between heartbeats, enabling accurate classification of multiple arrhythmia classes. Finally, fully connected dense layers and a Softmax classifier generate probability-based predictions for each heartbeat category.

The performance of the proposed model is evaluated using metrics such as accuracy, precision, recall, and F1-score, along with visualization techniques including confusion matrix and learning curves. The results demonstrate that the hybrid architecture effectively combines spatial and temporal learning to achieve reliable ECG classification. This system can assist clinicians in faster and more accurate diagnosis, contributing to intelligent healthcare solutions and early detection of cardiac abnormalities.

ECG BASED ARRHYTHMIA DETECTION

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Objective

The objective of this project is to develop an automated system for detecting and classifying cardiac arrhythmias using ECG signals.It utilizes a hybrid deep learning model for effective feature extraction.The system aims to improve classification accuracy and reduce dependency on manual analysis.It also supports early detection of heart abnormalities and assists in clinical decision-making.

Select domain (s) where your solution can be implemented

- Healthcare

Describe how your solution is going to meet the program outcomes

The project applies engineering knowledge and deep learning techniques to solve real-world healthcare problems.
It demonstrates problem analysis by addressing limitations of manual ECG interpretation through automation.The solution contributes to society by supporting early detection of heart abnormalities and improving healthcare decisions.

Describe the engineering solution

The proposed solution develops an automated ECG-based arrhythmia detection system using a hybrid deep learning model.ECG signals are preprocessed and analyzed using CNN and residual blocks to extract important features.These features are further processed using BiLSTM to capture temporal patterns in heart signals.The system classifies heartbeats into different arrhythmia categories, improving accuracy and supporting early diagnosis.

End Users / Stake holders of solution

End users include doctors, hospitals, patients, and researchers who benefit from accurate and automated ECG analysis.

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1.INTRODUCTION

1.1 PROJECT OVERVIEW

This project is designed to implement an automated system for detecting and classifying heart arrhythmias using ECG (electrocardiogram) signals with advanced deep learning techniques. Heart rhythm abnormalities are often difficult to identify manually, especially in large datasets, which creates a need for automated and accurate diagnostic solutions.

In this work, a hybrid deep learning model is designed by integrating Convolutional Neural Networks (CNN), BR-SquareNet residual learning blocks, and Bidirectional Long Short-Term Memory (BiLSTM) networks. The CNN component is responsible for extracting essential local features from ECG signals, while the residual blocks enhance these features by learning refined representations and improving training stability. The BiLSTM layer is used to capture sequential and temporal dependencies in heartbeat signals, enabling better understanding of rhythm variations.

The system is trained using the MIT-BIH Arrhythmia dataset, which contains annotated ECG heartbeat samples. Preprocessing steps include organizing and reshaping the data into a suitable format for model input. The performance of the model is evaluated using standard metrics such as accuracy, precision, recall, and F1-score to ensure reliable classification.

The proposed system demonstrates the capability of deep learning in handling complex biomedical signals and provides an efficient approach for automatic arrhythmia detection. This can support medical professionals by reducing diagnosis time and improving accuracy in clinical decision-making.

1.2 PURPOSE

The purpose of this project is to develop an automated and reliable system for detecting and classifying cardiac arrhythmias using ECG signals through deep learning techniques. The system aims to reduce the dependency on manual analysis, improve diagnostic accuracy, and assist healthcare professionals in identifying heart abnormalities at an early stage. By leveraging advanced neural network architectures, the project focuses on providing a fast, efficient, and scalable solution for real-time cardiac monitoring and decision support.

1.3 OBJECTIVE

The primary objective of this project is to design an intelligent system for automatic classification of ECG signals.It aims to apply advanced deep learning techniques to identify different types of heart arrhythmias.The model focuses on learning important patterns and rhythm variations from ECG data.It also aims to enhance prediction performance using a hybrid architecture.The overall goal is to provide a reliable tool for assisting in early and accurate heart disease diagnosis.

1.4 LITERATURE SURVEY

To address the limitations in existing research, the proposed approach focuses on applying Deep Learning techniques for accurate detection of arrhythmias from ECG signals. The study builds a strong understanding from basic ECG signal processing to model development, ensuring reliable and efficient results. The following section summarizes the existing research works. These existing works are analyzed in this section to understand their methodologies, strengths, and existing limitations.

[1] Detection and Classification of Electrocardiography using Hybrid Deep Learning Models (2025)

This study proposed a hybrid deep learning model combining CNN and LSTM for ECG signal classification. It focused on improving accuracy and F1-score by extracting both spatial and temporal features. However, the model faced challenges such as overfitting and limited availability of labeled data.

[2] Deep CNN-Based Detection of Cardiac Rhythm Disorders using PPG (2025)

This work utilized deep convolutional neural networks on PPG signals obtained from wearable devices to detect cardiac abnormalities. The model achieved good accuracy and sensitivity, but the performance was affected by noise present in wearable signal data.

[3] Smart Arrhythmia Detection using Single Lead ECG and Hybrid DNN Model (2025)

This research developed a hybrid deep neural network for arrhythmia detection using single-lead ECG signals. Although it showed promising accuracy and sensitivity, the limitation of using only a single lead reduced the robustness of detection.

[4] ECG-Based Heart Arrhythmia Classification using Feature Engineering and Hybrid ML (2025)

This work combined feature engineering with hybrid machine learning techniques for arrhythmia classification. It achieved good accuracy and recall, but heavily depended on manual feature extraction and dataset quality.

[5] ECG Analysis using Self-Attention-Based Autoencoder (2025)

This study applied a self-attention autoencoder model to improve ECG signal analysis. The model enhanced accuracy and precision but required a large amount of training data for effective performance.

[6] ACC Expert Consensus on Arrhythmia Monitoring after Stroke (2025)
This work presented clinical guidelines for arrhythmia monitoring in post-stroke patients. It focused on clinical applicability and decision pathways but required further validation for real-world implementation.

[7] Explainable Deep Learning for ECG-Based Arrhythmia Detection (2025)
This research introduced explainable AI techniques for ECG classification to improve interpretability. While it enhanced transparency, it faced challenges in balancing explainability and model complexity.

[8] The Role of AI in Cardiovascular Care (2025)
This study reviewed various AI applications in cardiology and highlighted the growing importance of intelligent systems in healthcare. However, it emphasized the need for regulatory frameworks and standardization.

[9] ECG-Based Arrhythmia Detection using Ensemble Learning (2024)
This work proposed an ensemble learning approach combining spatial, temporal, and long-range dependency models. It achieved high accuracy and sensitivity but suffered from computational complexity and data imbalance issues.

[10] Performance Comparison of In-Hospital Patient Monitoring Systems for Ventricular Arrhythmias (2024)
This study compared multiple in-hospital monitoring systems for detecting ventricular arrhythmias. It evaluated performance using detection rate and false alarm rate as key metrics. Although the systems showed effective detection capabilities, the study was limited by a small dataset and issues related to generalizability across different clinical environments.

1.5 REQUIREMENTS SPECIFICATION

1.5.1 HARDWARE SPECIFICATIONS

1.5.1.1 PROCESSOR

The basic job of a processor is to receive input and provide the appropriate output. While this may seem like a simple task, modern processors can handle trillions of calculations per second. The processor we used is i5.

1.5.1.2 RAM

RAM (Random Access Memory) is the hardware in a computing device where the operating system, application programs and data in current use are kept so they can be quickly reached by the device's processor. RAM is the main memory in a computer, and it is much faster to read data from RAM. The Ram recommended is 8GB.

1.5.1.3 HARD DISK

A hard disk drive (HDD) is a non-volatile computer storage device containing magnetic disks or platters rotating at high speeds.The hard disk is used as secondary storage for storing ECG datasets, trained models, and output results. A minimum of 256GB SSD or 500GB HDD is recommended to ensure sufficient space for data handling and smooth system performance.

1.5.2 SOFTWARE SPECIFICATIONS

1.5.2.1 OPERATING SYSTEM

An Operating System (OS) is an interface between computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling

input and output, and controlling peripheral devices such as disk drives and printers. Some popular operating systems include Linux Operating System, Windows Operating System etc. We have used Windows OS.

1.5.2.2 PYTHON PROGRAMMING LANGUAGE

- Python is a general-purpose programming language started by Guido van Rossum, which became very popular in short time mainly because of its simplicity and code readability. It enables the programmer to express his ideas in fewer lines of code without reducing any readability. We have used Python 3.6.5 version.

- **Python is Interpreted**
  Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP

- **Python is Interactive** − you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

- **Python is Object-Oriented** − Python supports Object-Oriented style or technique of programming that encapsulates code within objects.

- **Python is a Beginner's Language** − Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

LIBRARIES USED:

1. **Scikit-Learn**

   Scikit-Learn is probably the most useful library for machine learning in Python. It is on NumPy, SciPy and matplotlib, this library contains a lot of efficient tools for machine learning and statistical modelling including classification, regression, clustering and dimensionality reduction.

Scikit-Learn is characterized by a clean, uniform, and streamlined API, as well as by very useful and complete online documentation. A benefit of this uniformity is that once you understand the basic use and syntax of Scikit-Learn for one type of model, switching to a new model or algorithm is very straightforward.

2. Matplotlib

Matplotlib is a comprehensive library for creating static, animated and interactive visualizations. Matplotlib produces publication-quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and IPython shell, web application servers, and various graphical user interface toolkits.

3. NumPy

NumPy, which stands for Numerical Python, is a library consisting of multidimensional array objects and a collection of routines for processing those arrays. Using NumPy, mathematical and logical operations on arrays can be performed.

Using NumPy, a developer can perform the following operations –

- Mathematical and logical operations on arrays.
- Fourier transforms and routines for shape manipulation.
- Operations related to linear algebra. NumPy has in-built functions for linear algebra and random number generation.

4. Pandas

- A fast and efficient Data Frame object for data manipulation with integrated indexing.
- Tool for reading and writing between in-memory data structures and different formats: CSV and text files.
- Columns can be inserted and deleted from data structures for size mutability.
- Python with pandas is in use in a wide variety of academic and commercial domains, including Finance, Neuroscience, Economics, Statistics, Advertising, Web Analytics etc.

5. Seaborn

- Seaborn is a library for making statistical graphics in Python. It builds on top of matplotlib and integrates closely with pandas data structures.
- Seaborn helps you explore and understand your data. Its plotting functions operate on data frames and arrays containing whole datasets and internally perform the necessary semantic mapping and statistical aggregation to produce informative plots.
- Its dataset-oriented, declarative API lets you focus on what the different elements of your plots mean, rather than on the details of how to draw them.

6. WFDB (WaveForm DataBase Toolbox)

WFDB is a specialized library used for reading and processing physiological signal data such as ECG. It is commonly used with datasets like MIT-BIH to access signal records and annotations. In this project, WFDB is used to extract ECG signals and corresponding labels from raw files.

2. SYSTEM STUDY AND ANALYSIS

2.1 EXISTING SYSTEM

- The existing system mainly relies on manual analysis of ECG signals by medical experts. This process is time-consuming and requires specialized knowledge, making it less efficient for large-scale analysis.
- Traditional machine learning methods are used, which depend on manual feature extraction. This limits the ability to capture complex patterns and reduces overall accuracy.
- These systems have limited capability in handling real-time data and large datasets. They may also suffer from lower performance and lack scalability in practical applications.

2.2 PROPOSED SYSTEM

- The proposed system uses a hybrid deep learning model combining CNN, BR-SquareNet, and BiLSTM. This enables automatic feature extraction and improves the model's ability to learn complex ECG patterns.
- It captures both spatial features (waveform structure) and temporal dependencies (rhythm patterns), leading to more accurate arrhythmia classification.
- The system is efficient, scalable, and suitable for real-time applications. It reduces human effort and supports early detection and clinical decision-making.

2.3 THEORETICAL BACKGROUND

2.3.1 Deep Learning

Deep learning is a subset of machine learning that uses artificial neural networks to learn patterns from large amounts of data. Unlike traditional methods, it automatically extracts features from raw data without manual intervention. It is widely used in applications such as image processing, speech recognition, and biomedical signal analysis.

2.3.2 ECG Signal Analysis

An Electrocardiogram (ECG) represents the electrical activity of the heart over time. It consists of important components such as the P wave, QRS complex, and T wave. Variations in these waveforms help in identifying different types of cardiac arrhythmias, making ECG analysis essential for heart disease diagnosis.

2.3.3 Convolutional Neural Networks (CNN)

CNN is a deep learning technique used for extracting spatial features from data. In this project, Conv1D layers are applied to ECG signals to detect local patterns such as peaks and waveform shapes. These layers help in automatically learning important features from the signal.

2.3.4 Residual Learning

Residual learning introduces shortcut connections that allow the input to be added directly to the output of a layer. This helps in training deeper networks by preventing vanishing gradient problems. It also improves feature refinement and model performance.

2.3.5 Bidirectional Long Short-Term Memory (BiLSTM)

BiLSTM is a type of recurrent neural network that processes sequential data in both forward and backward directions. Since ECG signals are time-series data, BiLSTM helps in capturing temporal dependencies and rhythm patterns effectively.

2.3.6 Softmax Classification

Softmax is used in the final layer of the model to convert output values into probabilities for multiple classes. It helps in determining the most probable class for each ECG signal.

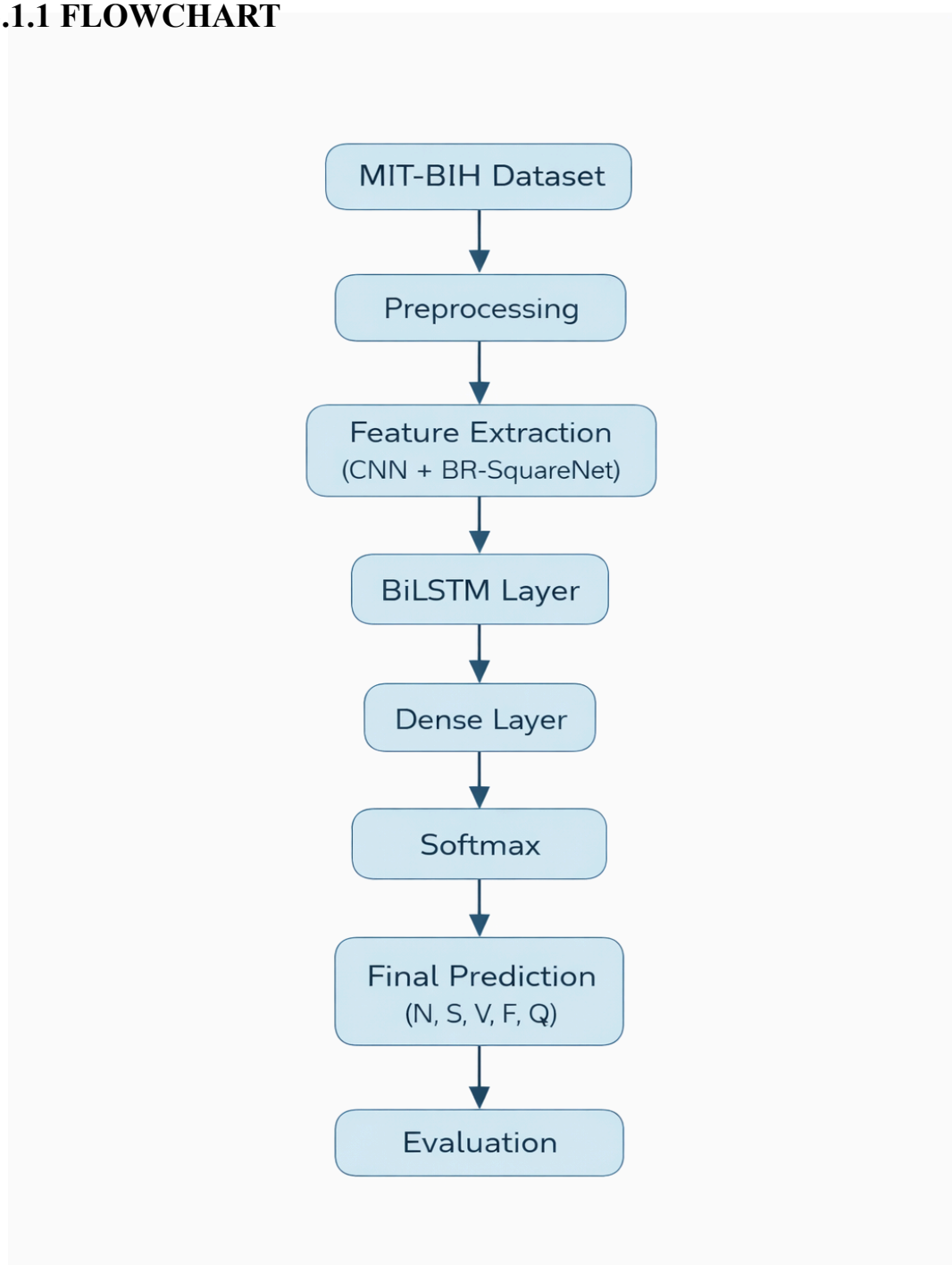
2.3.7 Performance Metrics

The model performance is evaluated using accuracy, precision, recall, and F1-score. These metrics help in measuring how well the model classifies ECG signals and detects abnormal heartbeats.

3. SYSTEM DESIGN

3.1 PROCESS

3.1.1 FLOWCHART



The system processes ECG data through preprocessing, feature extraction using CNN and residual learning, followed by BiLSTM for temporal analysis and classification using Softmax, with evaluation based on standard performance metrics.

Fig 3.1 Process flow chart

4. CODING MODULE

4.1 DATASET

Dataset Description

The dataset used in this project is the MIT-BIH Arrhythmia dataset, which is a standard benchmark dataset for ECG signal analysis. It contains annotated electrocardiogram (ECG) recordings collected from multiple patients, representing different types of heart rhythms.

Each record in the dataset is segmented into individual heartbeats, where each sample consists of approximately 280 time-step values representing the ECG waveform. The dataset includes multiple heartbeat classes such as Normal (N), Supraventricular (S), Ventricular (V), Fusion (F), and Unknown (Q), enabling multi-class classification.

Dataset Records :

Heart beats	Records	Patient, Age	Nbr. of beats	Total number of beats
Normal (N)	MIT-119	Female, 51	1543	8860
	MIT-200	Male, 64	1743	
	MIT-209	Male, 62	2620	
	MIT-212	Female, 32	923	
	MIT-221	Male, 83	2031	
Premature ventricular contraction (PVC)	MIT-119	Female, 51	444	2750
	MIT-200	Male, 64	823	
	MIT-214	Male, 53	256	
	MIT-221	Male, 83	396	
	MIT-233	Male, 57	831	
Left Bundle Branch block (LBBB)	MIT-109	Female, 64	2492	6610
	MIT-111	Female, 47	2115	
	MIT-214	Male, 53	2003	
Right Bundle Branch block (RBBB)	MIT-118	Male, 69	2163	7170
	MIT-124	Male, 77	1531	
	MIT-212	Female, 32	1825	
	MIT-231	Female, 72	1254	
	MIT-232	Female, 76	397	
Atrial Premature Contraction (APC)	MIT-118	Male, 69	96	1890
	MIT-200	Male, 64	30	
	MIT-209	Male, 62	383	
	MIT-232	Female, 76	1381	

4.1 Dataset Records

5. TESTING

5.1.Classification results of various models

After Testing all the classification models the Accuracy,Validation accuracy and Test scores obtained as the following:

Models	**Accuracy**	**Validation accuracy**	**Test**
CNN+BI-LSTM +BR-SquareNet	**99.87**	**96.44**	**96.53**
CNN	97.08	94.23	94.86
BI-LSTM	98.42	94.19	95.06
BR-SquareNet	99.42	94.80	95.42
CNN+BI-LSTM	99.24	96.56	96.34
CNN+BR-SquareNet	99.47	94.60	94.62
BI-LSTM+BR-SquareNet	99.19	94.89	94.63

Table 5.1: Performance Comparison of Different Models

The performance of different models is compared using accuracy, validation accuracy, and test accuracy. Among all models, the proposed hybrid model (CNN + Bi-LSTM + BR-SquareNet) achieved the highest performance with 99.87% training accuracy, 96.44% validation accuracy, and 96.53% test accuracy.

The hybrid model (CNN + Bi-LSTM + BR-SquareNet) achieved the best performance compared to individual and partial hybrid models, proving the effectiveness of combining spatial, refined, and temporal feature learning.

6. CONCLUSION

6.1 CONCLUSION AND FUTURE WORK

The project successfully developed an automated ECG-based arrhythmia detection system using a hybrid deep learning model. It effectively combines CNN, residual learning, and BiLSTM to achieve accurate classification. The system improves efficiency and supports early diagnosis of heart abnormalities.The model demonstrates strong capability in handling complex ECG patterns and improving diagnostic reliability. It also reduces the dependency on manual analysis, making the system more efficient and scalable for real-world applications.

The system can be enhanced by using larger datasets and integrating real-time ECG monitoring devices. Advanced models and explainable AI can be applied to improve performance and interpretability. It can also be deployed as a mobile or web application for practical use.

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